CharAnalysis 2.0: June 2026 Update

SNI warning · Chronological uncertainty ensemble

GitHub “dev” branch:

- NEWS https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis_2_0_R/NEWS.md
- Chronological Uncertainty Vignette:

  https://github.com/phiguera/CharAnalysis/blob/dev/CharAnalysis_2_0_R/vignettes/chronological_uncertainty_draft.md

Phil Higuera — 25 June, 2026

Today's updates

1 SNI warning

When SNI falls below 3.0, *CharAnalysis* now flags it explicitly in the console.

The vignette's recommended workflow was rewritten to position SNI evaluation as Step 2, before any fire-history interpretation.

2 Chronological uncertainty

New ensemble pipeline: runs *CharAnalysis* 1,000× across MCMC chronologies, propagating age uncertainty through all analytical steps.

Reports per-peak:

- Detection frequency
- Timing uncertainty (95% CI)
- Ensemble-only candidate peaks.

Examples from CO, CH10, and SI17.

1. SNI warning:

Before
SNI was computed and plotted (Fig 4), but no message appeared if it fell below the threshold of 3.0. Users could overlook low-SNI intervals.

After
CharAnalysis now emits a styled red advisory at the end of all console output when SNI < 3.0.

Local thresholds: reports % below threshold

Global threshold: reports the scalar SNI value

```
R Console

CAUTION
✕ 12.8% of samples have an SNI < 3.0,
the minimum value recommended by Kelly
et al. (2011) to identify records
suitable for peak detection. Carefully
consider whether and how to interpret
portions of this record with SNI < 3.0.

ℹ Run 'char_plot_sni(out)' to visualize
the SNI series and identify which
portion(s) of the record fall below the
threshold.
```

Vignette: SNI evaluation before interpretation

1

Background smoothing

Figs 1 & 3: check that $C_{background}$ tracks low-frequency variation without absorbing peaks. Adjust smoothing window as needed.

2

Evaluate signal quality (SNI)

Fig 4: inspect SNI before any fire-history results. If SNI < 3 for substantial portions, peak detection in those intervals is not well justified.

3

Threshold diagnostics

Fig 2: examine fit of noise model to C_{peak} distribution. Most useful when Fig 4 suggests instability.

4

Analytical figures

Figs 6–9: zone comparisons, cumulative peaks, FRI distributions, fire history. Only after diagnostics are acceptable.

2. Chronological uncertainty

***CharAnalysis* alone does not propagate uncertainty from an age-depth model into peak detection** because it uses one age-depth model (e.g., median chronology).

Bayesian age-depth models (rbacon, MCAgeDepth) generate thousands of plausible chronologies via MCMC sampling. Each shifts sample ages and thus accumulation rates, CHAR, thresholds, and peak detection outcomes.

This tool integrates chronological uncertainty to address three questions:

1. How consistently is each peak identified?
2. What is the range of ages assigned to specific peaks?
3. Are there peaks common in the ensemble but absent from the main *CharAnalysis* run?

Most valuable for:

- Identifying & timing individual peaks (e.g., for comparison to other events)
- Records with meaningful age-model uncertainty

Less critical for:

- Fire-regime statistics (mFRI, fire frequency), because timing shifts among peaks tend to covary.

Ensemble workflow: four steps

1

Build age-depth model

(external: rbacon / MCAgeDepth)

Extract n = 1,000 chronologies from the MCMC output, preserving temporal covariance.

Any age-depth tool works; rbacon is the primary target.

2

Run the ensemble

(char_run_ensemble.R)

CharAnalysis pipeline runs once per chronology.

Parallel processing: ~2 min for 1,000 iterations on a modern desktop (19 cores).

3

Analyze ensemble results

(run_ensemble_analysis.R)

Use adaptive matching window for each reference peak.

Reports:

- Detection frequency
- 95% CI on timing
- Ensemble-only peaks (10% threshold).

4

Visualize

(plot_ensemble_figure.R)

Four-panel figure:

- CHAR + benchmark
- Detection frequency + peak types
- SNI
- Age-depth uncertainty ribbon.

Adaptive matching window

For each benchmark peak k, the matching window defines how far an ensemble detection can lie and still be attributed to the same fire event:

$$\mathrm{window}_k = \max(\mathrm{mFRI}_z/2,\ \mathrm{CI}_{95,k}/2)$$

mFRI_z - mean FRI for the zone containing peak k
$\mathrm{CI}_{95,k}$ - 95% width of chronological uncertainty at peak k

$\mathrm{mFRI}_z/2$: once a peak shifts > half the mean FRI, it is considered to be a different event

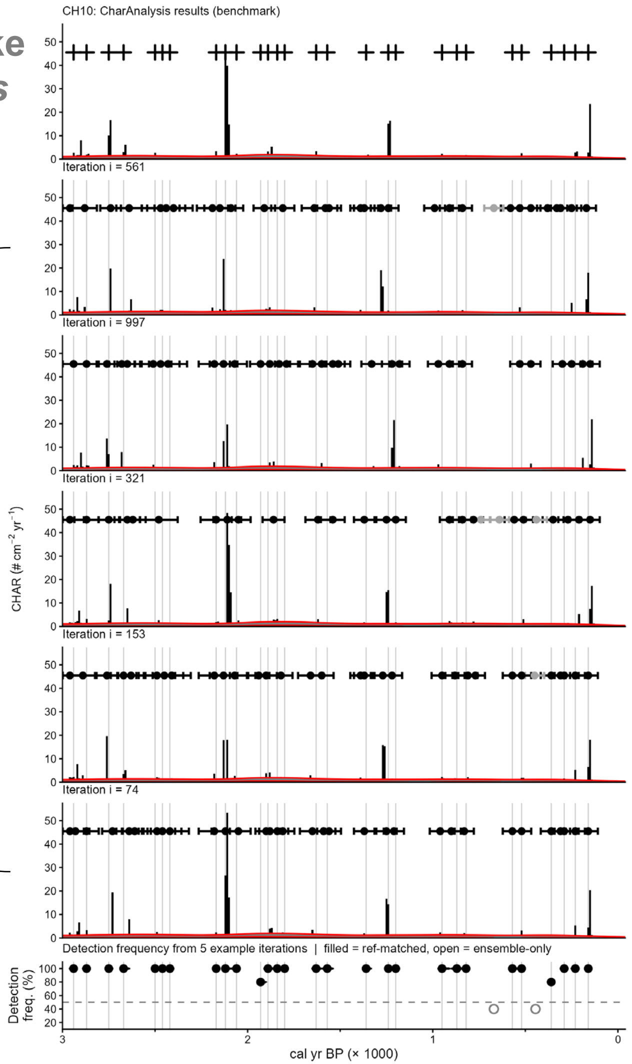
$\mathrm{CI}_{95,k}/2$: expands window where chronology is poorly constrained. Ensemble detections of the same peak, shifted by an uncertain chronology, are still correctly attributed

In practice, **$\mathrm{CI}_{95,k}/2$ should define windows in most records;** $\mathrm{mFRI}_z/2$ is a min. bound in well-constrained records (e.g., CH10).

Chickaree Lake
CharAnalysis
(benchmark)

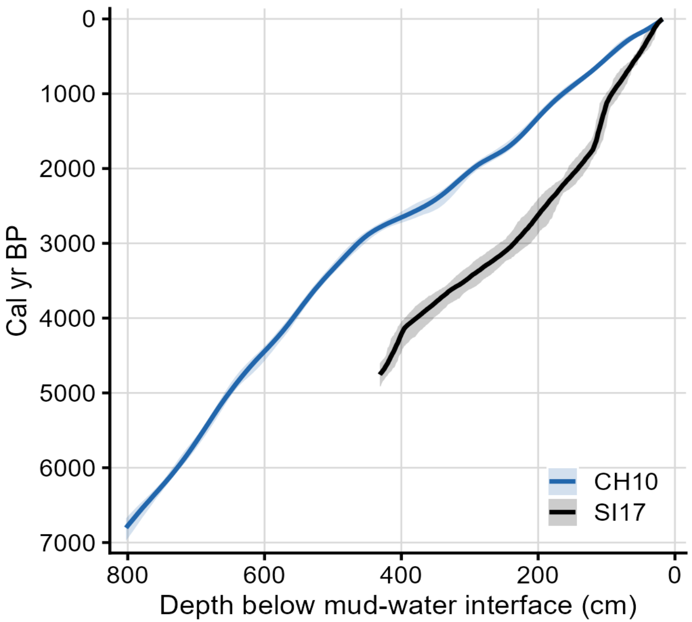
Iteration i,
with search
windows

Matched peaks (black)
Unmatched (white)



Site examples: CH10 and SI17

	CH10 (Chickaree Lake, CO)	SI17 (Silver Lake, MT)
Chronological control	Unusually well-dated (25 ^{14}C dates, ^{210}Pb)	Fewer dates (15 ^{14}C, 3 tephra, 14 ^{210}Pb)
Age-depth model source	MCAgeDepth	Bacon
Benchmark peaks	59	25
Mean FRI	104 yr	193 yr
Matching window (median ±)	±72 yr	±162 yr
Detection freq (median)	99%	99%
Timing SD (median)	23 yr	53 yr
Ensemble-only peaks	9 (6 near-ref, 3 indep.)	0 (coherence filter)



Median chronology ± 95% CI. CH10: narrow, uniform. SI17: wider.

CH10 — ensemble results

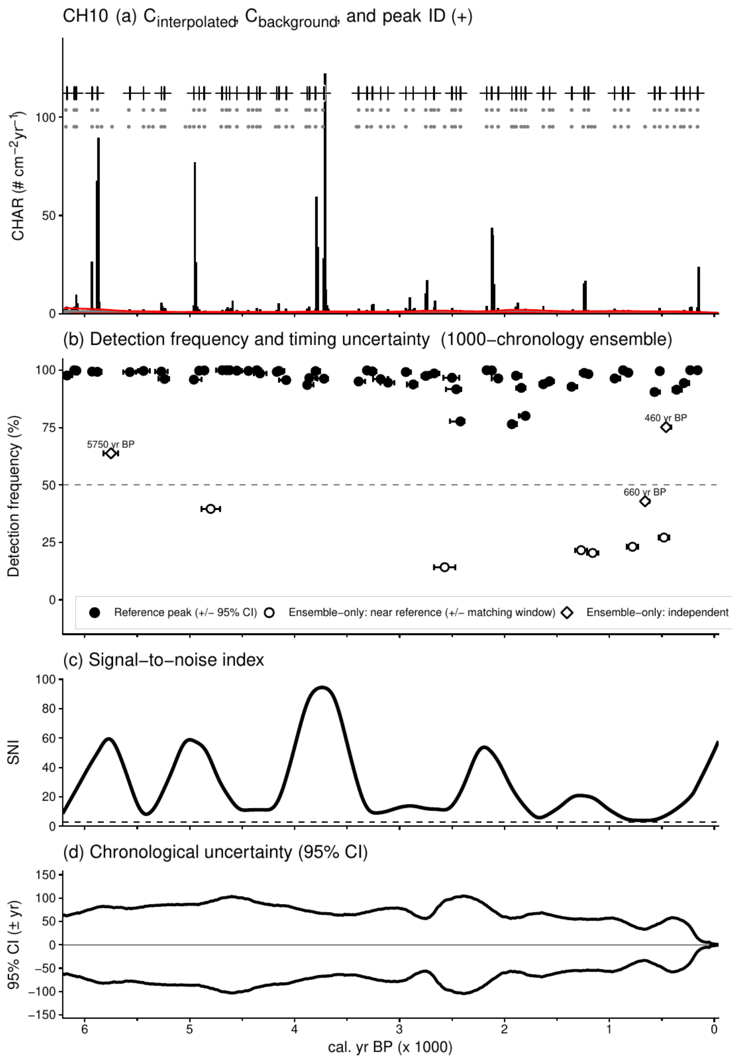
Benchmark:

Detection frequency, with 95% timing CI:

Reference peaks (black circles)
Ensemble-only: near reference (white circles)
Ensemble-only: independent (white diamonds)

SNI:

Age-depth 95% CI:



SI17 — ensemble results

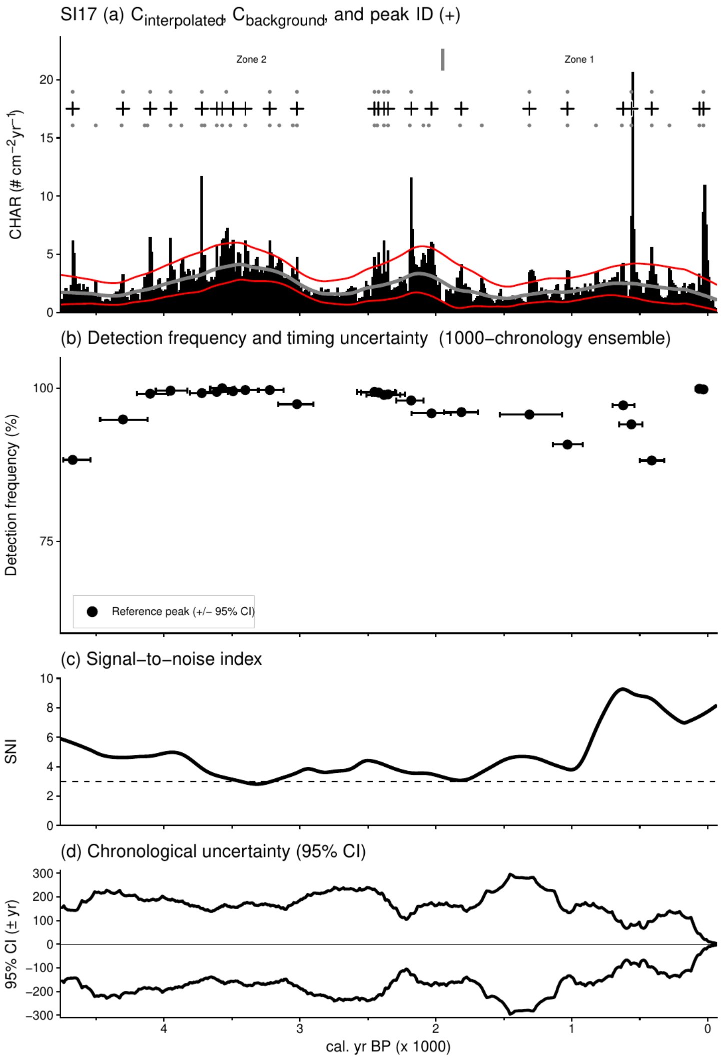
Benchmark:

Detection frequency, with 95% timing CI:

Reference peaks (black circles)
Ensemble-only: near reference (NONE)
Ensemble-only: independent (NONE)

SNI:

Age-depth 95% CI:



Status & next steps

Done

- SNI < 3.0 advisory in console output and added as Step 2 in recommended workflow (vignette)
- Figure reorganization: diagnostic (1–5) vs. analytical (6–9)
- Background sensitivity function added to R, to match Matlab version
- Chronological uncertainty working version

Up next

- Package char_run_ensemble() as exported function with Roxygen docs
- Add rbacon / rplum interface
- Write chronological_uncertainty.Rmd vignette
- Apply ensemble to additional sites beyond CH10 and SI17
- Package dev work as v2.0.4 (SNI advisory, figure reorganization, age-uncertainty function)